

# ECE 268 Project Progress Report

Lightweight Block Ciphers for Constrained Devices: PRESENT-80 and SPECK-64

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## Project Summary

**Goal:** Implement PRESENT-80 and SPECK-64 from scratch with encryption, decryption, and key schedules. Each cipher will be wrapped in CBC and CTR modes, validated against published test vectors, and benchmarked on GPU against a sequential CPU baseline and AES-128, with IoT/RFID as the target setting where AES is too expensive.

**Status:** PRESENT-80 is complete on both CPU and GPU. All four official test vectors pass and GPU performance has been measured. SPECK-64, CBC/CTR modes, and the AES baseline are pending.

## 1 Work Completed So Far

### Implementations

- **Repository:** Current files are `present80.cpp`, `present80.cu`, and `CMakeLists.txt`. The planned files `speck64.cpp`, `speck64.cu`, `modes.cpp`, and an AES baseline are not yet written.
- **PRESENT-80 CPU:** 64-bit block, 80-bit key, 32 rounds per Bogdanov et al. The key schedule rotates the 80-bit register left by 61, applies the S-box to the top nibble, and XORs the round counter into bits 19–15. Encryption runs 31 full rounds of `addRoundKey`, `sBoxLayer`, and `pLayer`, then adds the final round key. Decryption reverses those steps using `INV_SBOX` and the inverse P-layer. The bit permutation formula is  $p(j) = (16j) \bmod 63$  with bit 63 fixed in place.
- **PRESENT-80 GPU:** A CUDA port of the CPU implementation. Round keys and S-box tables go into `__constant__` memory so they are broadcast across each warp without extra traffic. The encrypt and decrypt kernels each assign one thread per cipher block. The P-layer helper is `__device__ __forceinline__` with `#pragma unroll` on the 63-iteration loop. Key expansion runs on the CPU and the result is copied to constant memory before the kernel launches. The test bench encrypts and then decrypts  $2^{20}$  distinct blocks and verifies that every output matches the original plaintext.
- **Build system:** `CMakeLists.txt` uses a `USE_CUDA` flag. `ON` builds `present80.cu` with `nvcc` targeting the native GPU architecture. `OFF` builds `present80.cpp` as a C++17 binary with `-O4`.

### Experiments and Evaluations

- **Setup:** Test vectors are the four canonical cases from Bogdanov et al. GPU timing uses CUDA events over  $2^{20}$  blocks on an NVIDIA RTX 4070 Mobile running CUDA 12.0. CPU timing uses `std::chrono::high_resolution_clock` on the same machine at  $\approx 3.8$  GHz.
- **Completed:** All four PRESENT-80 spec vectors pass on the CPU build. GPU round-trip correctness was verified across all  $2^{20}$  blocks. CPU and GPU encryption throughput were both measured on the same workload.
- **Pending:** SPECK-64 and CBC/CTR tests have not been run because those components are not yet implemented.

| Cipher / Platform          | Block/Key | Enc/Dec       | Spec Vectors | Throughput                   |
|----------------------------|-----------|---------------|--------------|------------------------------|
| PRESENT-80, CPU sequential | 64 / 80   | Round-trip OK | 4/4 Pass     | $\approx 0.64$ MB/s          |
| PRESENT-80, GPU RTX 4070M  | 64 / 80   | Round-trip OK | 4/4 Pass     | 1.07 GB/s enc, 1.43 GB/s dec |
| SPECK-64, CPU              | 64 / —    | —             | —            | Not started                  |
| SPECK-64, GPU              | 64 / —    | —             | —            | Not started                  |
| AES-128 baseline           | 128 / 128 | —             | —            | Pending                      |

## Preliminary Results

PRESENT-80 passes all published vectors. For the same  $2^{20}$ -block workload the GPU finishes in 7.85 ms versus 12,518 ms on the CPU, a roughly  $1595\times$  speedup. The CPU implementation is intentionally unoptimized and sits at  $\approx 4300$  cycles/byte.

## 2 Challenges and Issues Encountered

- **Key-schedule bit ordering:** Loading the 80-bit key into `std::bitset<80>` requires careful mapping because key byte 0 is the most significant byte. An off-by-one in the bit indexing produces wrong round keys that still pass the round-trip test, so the error only surfaces when running the official vectors.
- **Round key extraction:** The spec takes the leftmost 64 bits of the 80-bit register as the round key. Getting the MSB/LSB alignment right between the `bitset` and the `uint64_t` required cross-checking against each test vector.
- **P-layer boundary case:** The CPU version handles the  $j = 63$  fixed point with a ternary inside a 0-to-63 loop. The GPU port splits this into a 0-to-62 loop with a separate bit-63 check, which is easier to verify.
- **CMake and CUDA:** `CMAKE_CUDA_COMPILER` is not resolved automatically in all environments. The GPU build currently requires either the `CUDACXX` environment variable or calling `nvcc` directly.
- **Scope:** PRESENT-80 uses a 64-bit block with an 80-bit key and 32 rounds. For SPECK-64, the key width needs to be confirmed from the SIMON/SPECK appendix before implementation starts. Gate-count analysis is out of scope and the PRESENT reference value of 1570 GE will only be cited for context.

## 3 Plans for the Next Two Weeks

**Week 1 (May 22–28):** Implement SPECK-64 on CPU and GPU from the NSA spec. Validate against the published appendix vectors. Write CBC and CTR mode wrappers that work with both ciphers and verify round-trip correctness for each.

**Week 2 (May 29–June 4):** Add an AES-128 baseline using OpenSSL. Run benchmarks covering GB/s, cycles/byte, and key-schedule cost at 1 KB, 16 KB, and 64 KB payloads. Compile the comparison table and finish the report and slides.

**Planned evaluations:** Full vector sweep for SPECK-64 and both mode wrappers. Median GB/s over 10 GPU runs. GPU vs. CPU speedup. `.text` size comparison across PRESENT-80, SPECK-64, and AES-128. A Feistel vs. SPN trade-off discussion for constrained devices.

**Presentation (June 4):** 10–12 slides on cipher architectures, GPU kernel design choices, benchmark results, and lightweight-crypto analysis. Slides will be uploaded to the shared Drive before the session.

| Member          | Tasks                                                                    |
|-----------------|--------------------------------------------------------------------------|
| Devansh Gupta   | SPECK-64 CPU/GPU implementation, AES-128 baseline, results plots         |
| Priyansh Parikh | CBC/CTR mode wrappers, benchmark scripts, PRESENT-80 test-vector harness |
| Jeevan N V      | SPECK-64 vector validation, build cleanup, slides                        |

## Feasibility and Direction

PRESENT-80 is done and SPECK-64 is the immediate critical path. One to two days of implementation and one day of debugging is a reasonable estimate. CBC/CTR wrappers are straightforward once both ciphers pass vectors. If a custom CUDA AES takes too long, OpenSSL provides a ready-made reference for the baseline. The final deliverables are passing test vectors for both ciphers in all modes, a GPU vs. CPU comparison table, and a lightweight-crypto analysis for the presentation.

*References:* Bogdanov et al., PRESENT (CHES 2007). Beaulieu et al., SIMON/SPECK (ePrint 2013/404).